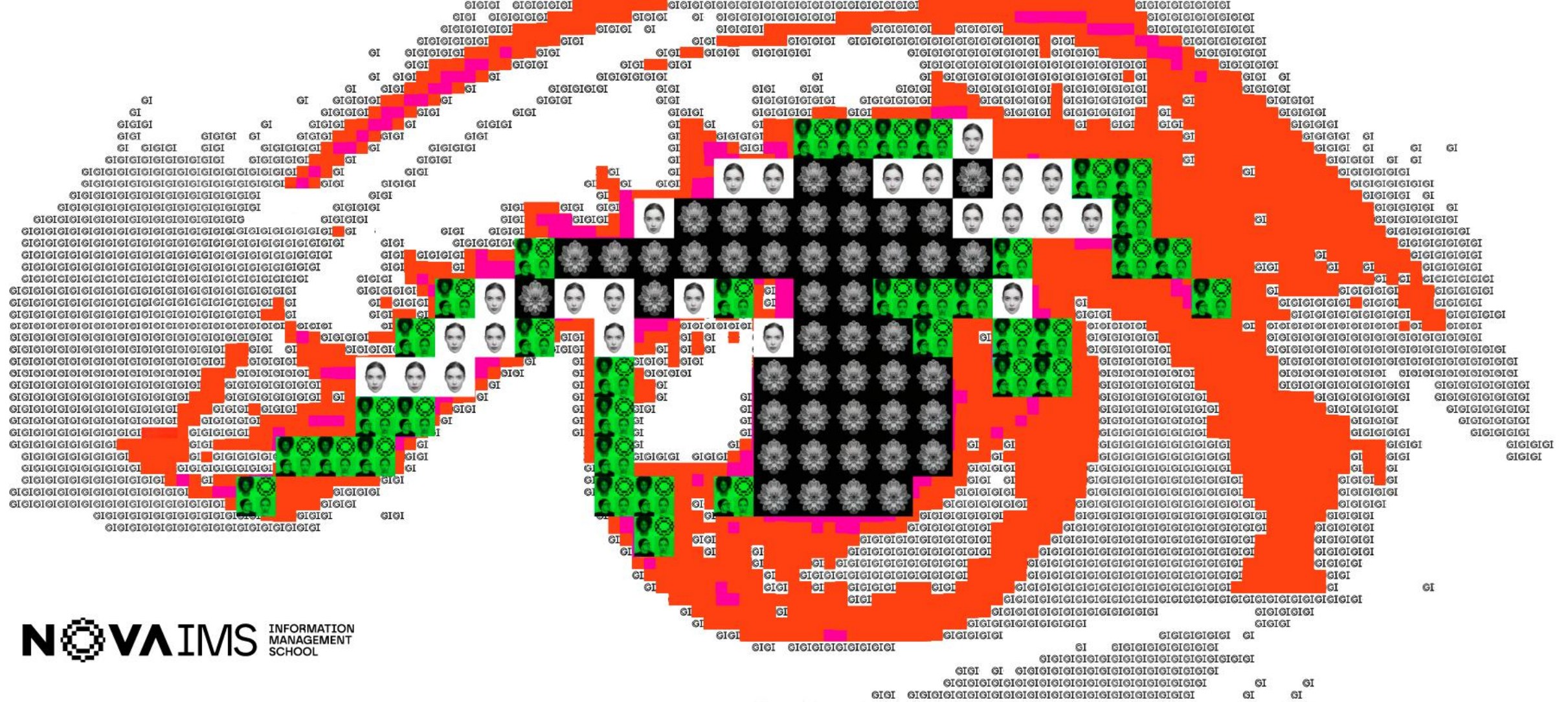




This course introduces the **foundations and applications of Natural Language Processing (NLP) methods**. By the end of the course, students will:

- Understand how **to prepare and represent text data** through key **preprocessing techniques** (e.g., tokenization, stemming, lemmatization) and **feature extraction methods** like **Bag of Words, TF-IDF, and word embeddings**.
- Apply traditional and **deep learning models** to **analyze and classify text**, including logistic regression, LSTMs, Transformers, and Large Language Models (LLMs).
- Explore traditional Text Mining/NLP tasks, including **sentiment analysis, document classification** and **topic modeling**.
- Gain hands-on **experience through lab sessions and a final project**, using tools such as scikit-learn, Hugging Face Transformers, and LangChain to **build practical NLP applications**, including chatbots.

Teachers:

Bruno Jardim (bjardim@novaims.unl.pt)

Ricardo Dona (rdona@novaims.unl.pt)

Catarina Nascimento (cnascimento@novaims.unl.pt)

Office hours:

Please see with your teacher, to get his office hours

Lecture & Labs topics (subject to review/change)

W1: Introduction to NLP

Introduction to text preprocessing and classification using Bag of Words models.

W2: Word Representations

Explore word vector representations (e.g., Word2Vec, GloVe) and apply them in classification.

W3: Sequential Models

Sequential models (RNNs, LSTMs) and applications in classification and sequence modeling tasks.

W4: Large Language Models I

Use embeddings from Transformer models to perform classify and cluster documents.

W5: Large Language Models II

Develop chatbots using LLMs and explore different tasks, like dialogue and sentiment analysis.

W6: Agentic AI

Design and implementation of agents, focusing on tool use, decision-making, and task orchestration.

W7: Invited Speaker

Guest lecture showcasing practical NLP and AI use cases and industry perspectives.

Google Colab: <https://colab.google/>



Grading, results from the following evaluation variables:

- a. **Project * (50%)**
- b. **Final Exam (50%)**

To pass a **minimum of 9,5** must be obtained in the Project or Final exam.

Project always accounts for 50% of the final grade, even in 2nd Call.

* Group composition defined in the Labs

Deadlines:

Project: **15 June 2026 at 23h59**

Exam 1: **29 June 2026 at 15h00**

Exam 2: **17 July 2026 at 16h00**

- Dan Jurafsky and James H. Martin. **Speech and Language Processing** 3rd ed.
<https://web.stanford.edu/~jurafsky/slp3/>
- Alammur, J., & Grootendorst, M. (2024). **Hands-On large language models: Language Understanding and Generation**. Sebastopol : O'Reilly Media.
NOVA IMS Access: <https://search.library.novaims.unl.pt/cgi-bin/koha/opac-detail.pl?biblionumber=97234>

Other reading materials:

- Materials distributed in each class during the term

Thank you!

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